

Why do GOOD PROGRAMMERS *fail*
at building GOOD SOFTWARE?





Project Management Concepts

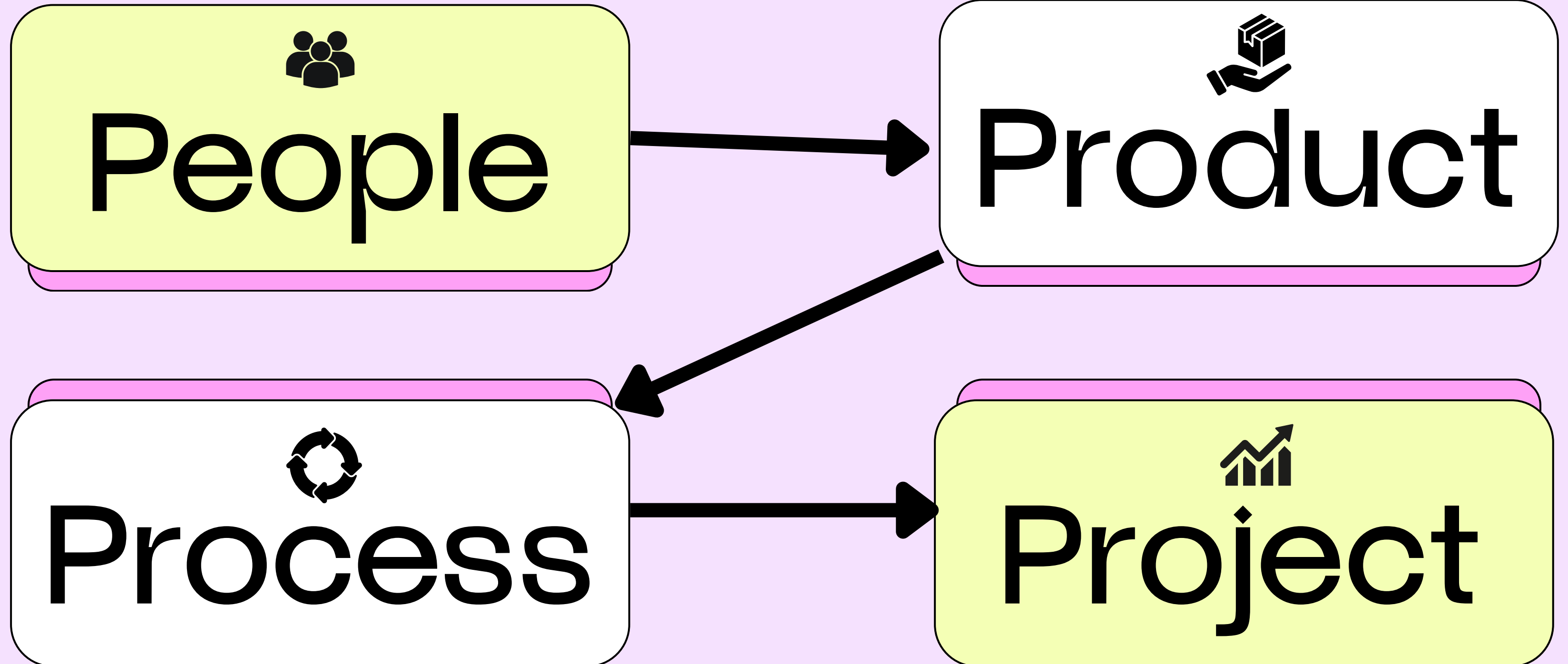


Chapter 24

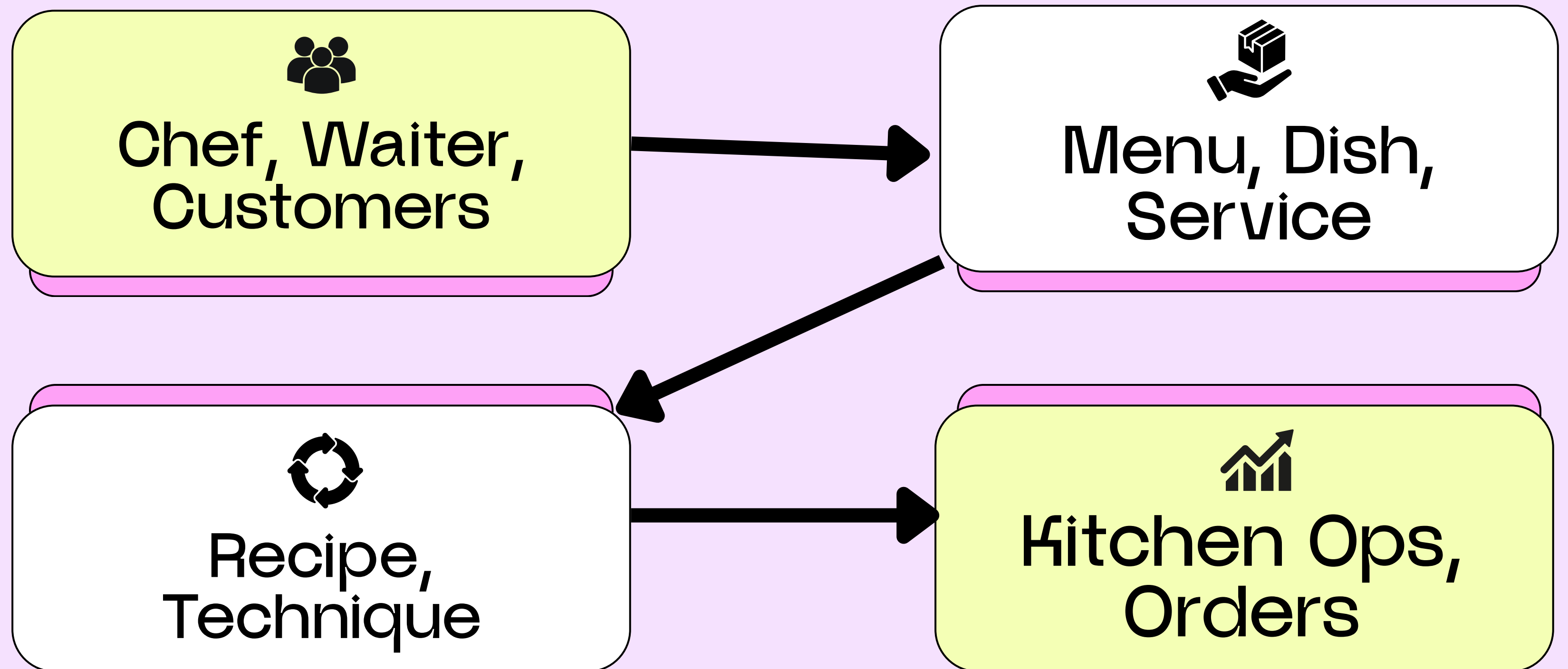


The Management Spectrum

REMEMBER THE 4 P's:



Restaurant Analogy



The People

People Over Tools

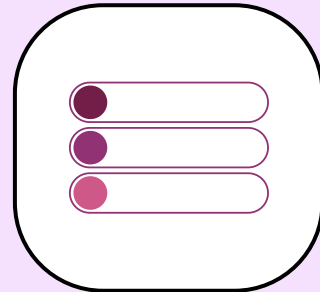
The Core Principle: "It's not the tools that we use, it's the people."
— IEEE Survey of Tech Execs

5 Key Stakeholder Constituencies



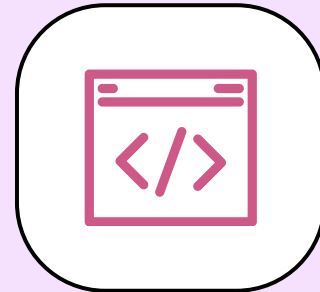
Senior Managers

Define business issues and strategic goals driving the software.



Project Managers

Plan, motivate, organize, and guide the practitioners.



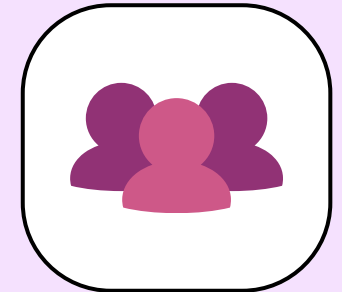
Practitioners

Deliver the technical skills needed to build the product.



Customers

Specify requirements and fund project outcomes.



End Users

Interact directly with the software in production.

Team Leadership and the MOI Model

Weinberg's MOI Leadership Model

Motivation: Encouraging technical staff to perform at their highest ability.

Organization: Structuring processes that turn ideas into working software.

Innovation: Fostering creativity within project bounds and constraints.

Key Managerial Traits

Problem Solving: Diagnosing organizational & technical issues quickly.

Managerial Identity: Taking charge while trusting team instincts.

Achievement: Rewarding initiative and controlled risk-taking.

Influence: Reading verbal/non-verbal signals in high-stress situations.

Constantine's 4 Team Paradigms

Closed Paradigm

Traditional top-down hierarchy. Works well for highly routine software similar to past efforts, but stifles innovation.

Random Paradigm

Loose structure driven by individual autonomy. Excellent for breakthroughs, but struggles with orderly delivery.

Open Paradigm

Collaborative and consensus-based. Combines control with innovation, ideal for solving complex problems.

Synchronous Paradigm

Compartmentalizes problems into independent modules, allowing team members to work in parallel with minimal meeting overhead.

Team Effectiveness: Jelled & Agile Teams

The "Jelled" Team Goal

A tightly knit group where the whole is greater than the sum of parts exhibiting organic momentum and high mutual trust.

Agile Mindset

Self-organizing teams with high local autonomy, small iterations, and daily informal collaboration.

5 Team Toxins to Avoid: Frenzy, frustration, fragmented process, role ambiguity, repeated failure.

The Product

Software Scope

The first software project management activity is the determination of a *software scope*, which is defined by answering the following questions.

1

Context

2

Information Objectives

3

Function and Performance

Software Scope

Context

- How does the software to be built fit into a larger system, product, or business context?
- What **constraints** are imposed as a result?

Information Objectives

- What customer-visible data objects are produced as **output** from the software?
- What data objects are required for **input**?

Function and Performance

- What **function** does the software perform to transform input data into output?
- Are any special performance characteristics to be addressed?

**Must be unambiguous and understandable at the management and technical level.*

Software Scope

**quantitative data must
be explicitly stated**

e.g. number of simultaneous users, target
environment, maximum allowable response time

**constraints and/or limitations
are noted**

e.g. product cost restricts memory size

mitigating factors

e.g., desired algorithms are well understood and
available in Java

SET YOUR BOUNDARIES !!

If you don't, people don't know when to stop.

Problem Decomposition

a.k.a. **Partitioning** or Problem Elaboration,
and is applied in two major areas.

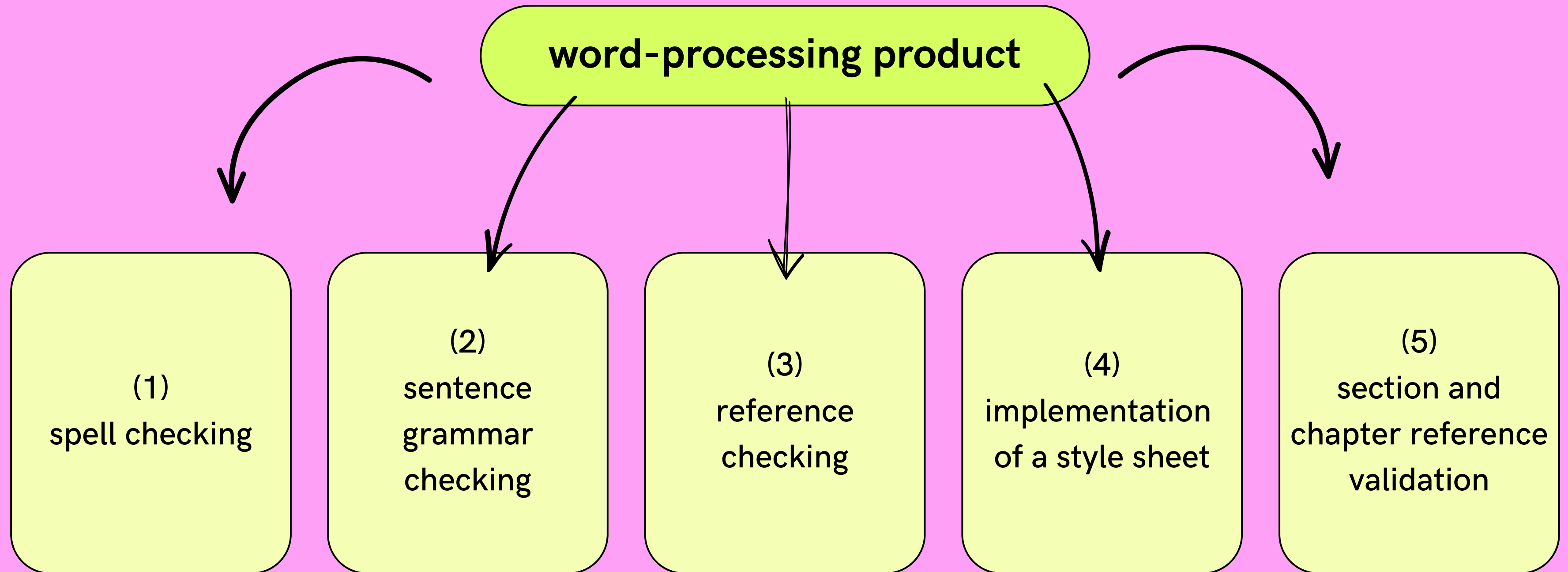
1

functionality and content (information)
that must be delivered

2

the process that will be
used to deliver it

Divide and Conquer



The Process

Selecting the Right Process

Framework Activities

- Communication
- Planning
- Modeling
 - Analysis of requirements
 - Design
- Construction
 - Code generation
 - Testing
- Deployment

The framework activities that characterize the software processes are applicable to all software projects.

Selecting the Right Process

The team must decide which model is most appropriate for:

1

Customers and the ones who will do the work.

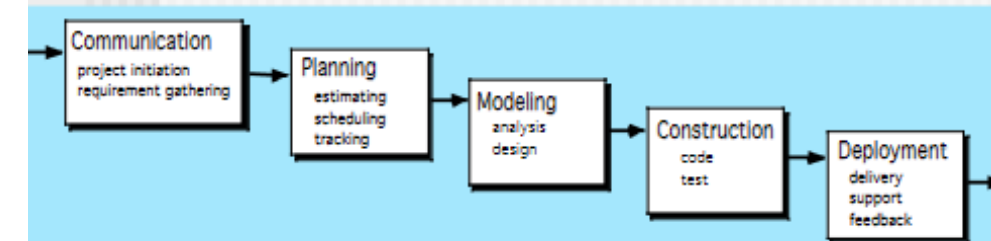
2

Characteristics of the product itself.

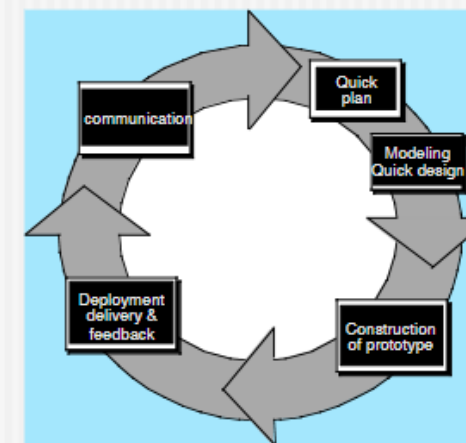
3

Project environment in which the team works.

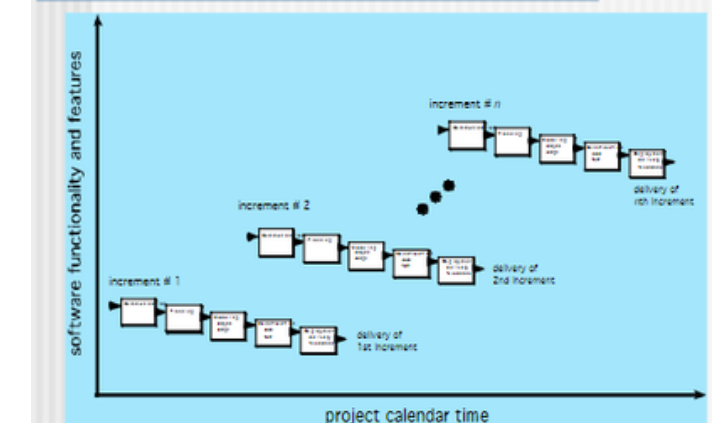
The Waterfall Model



Evolutionary Models: Prototyping



The Incremental Model



Melding the Product and Process

Project planning begins with the melding of the product and the process.

Mapping WHAT you need to build against HOW you build things.

COMMON PROCESS FRAMEWORK ACTIVITIES	<i>communication</i>	<i>planning</i>	<i>modeling</i>	<i>construction</i>	<i>deployment</i>
Software Engineering Tasks					
Product Functions					
Text input					
Editing and formatting					
Automatic copy edit					
Page layout capability					
Automatic indexing and TOC					
File management					
Document production					

Product Decomposition

PD commences when the project manager asks,
"How do we accomplish this framework activity?"

Communication

1

Develop list of clarification
issues.

2

Meet with stakeholders to
address clarification issues.

3

Jointly develop a statement
of scope.

4

Review the statement of scope
with all concerned.

5

Modify the statement of
scope as required.

The Project

10 Signs a Project is in Jeopardy

according to John Reel

1

Software people don't understand their customer's needs.

2

The product scope is poorly defined.

3

Changes are managed poorly.

4

The chosen technology changes.

5

Business needs change [or are ill defined].

Deadlines are unrealistic.

6

Users are resistant.

7

Sponsorship is lost [or was never properly obtained]

8

The project team lacks people with appropriate skills.

9

Managers [and practitioners] avoid best practices and lessons learned.

10

The 90-90 Rule

according to jaded industry professionals

The first 90% of a system absorbs 90% of the allotted effort and time.

The last 10% takes another 90% of the allotted effort and time

A 5-Part Commonsense Approach

according to John Reel



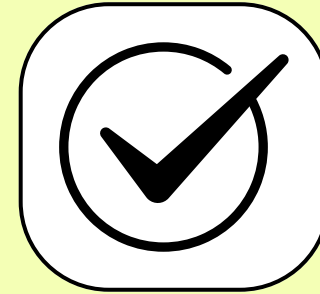
**Start on the
right foot.**



**Maintain
momentum.**



**Track
progress.**



**Make smart
decisions.**



**Conduct a
postmortem
analysis.**

The W⁵HH Principle

according to Barry Boehm

W

Why is the system being developed?

W

What will be done?

W

When will it be done?

W

Who is responsible for a function?

W

Where are they located organizationally?

H

How will the job be done technically and managerially?

H

How much of each resource is needed?

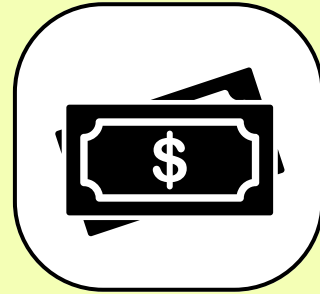
Critical Practices

according to the Airline Council



**Metric-based
project
management**

(Chapter 25)



**Empirical cost
and schedule
estimation**

(Chapter 26-27)



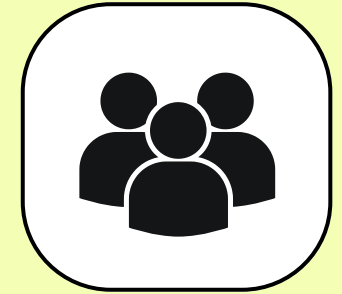
**Earned value
tracking**

(Chapter 27)



**Defect
tracking
against quality
targets**

(Chapter 14-16)



**People aware
management**

(Section 24.2)

Summary

Project management is an umbrella activity. It spans the entire software process, not just one phase.

People remain the most critical or pivotal factor.

Success depends on the 4 P's:

- **People:** organized, motivated, well-coordinated
- **Product:** clearly communicated and partitioned requirements
- **Process:** adapted to fit the people and product
- **Project:** structured to enable team success

**Thank
you!**



**Espejo
Fangonil
Salting**